volatility_filter

equity strategy | family: equity-volatility | status: research | derived from `balanced_leverage`

What this strategy does

Balanced Leverage, skipping long/short x jumpiness groups that have not made money so far

Why it should work. Skip the cells that history says do not pay -- but the data shows a gradient, not a bad cell.

Design notes (from the strategy's source file).

Strategy 9a / 9c -- walk-forward volatility-cell filter (alone and combined with inverse-vol).

Drops a (side, idio-vol tercile) cell from new entries when its own history says it has not earned its keep, judged ONLY on quarters whose positions had fully exited. Result: filtering does not help on its own (Sharpe 1.23 vs 1.30 baseline-equivalent run; the effect is a smooth gradient, not one broken cell) and adds nothing once inverse-vol sizing is on (9c 1.41 < 9b 1.47). Kept as a documented negative result: size by vol, do not filter by it.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

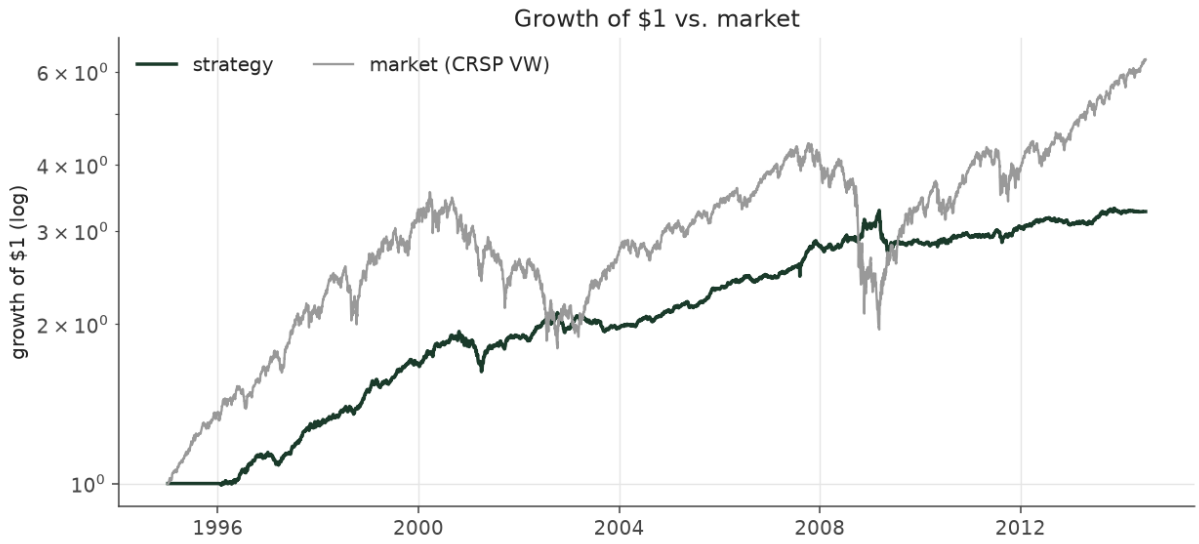
metric	volatility filter	market (CRSP VW)
Annualized return	6.27%	9.94%
Annualized volatility	5.05%	19.35%
Sharpe (NAV-pct)	1.23	0.59
Sortino	1.79	
Max drawdown	-16.7%	-55.5%
Calmar	0.38	
Final NAV ($10M start)	$32,665,477	
Worst day	-3.65%	
% positive days	51.8%	
Sharpe 1996-2001	1.90	
Sharpe 2002-2007	1.44	
Sharpe 2008-2013	0.48	

Correlation of daily returns with the market: **0.04**.

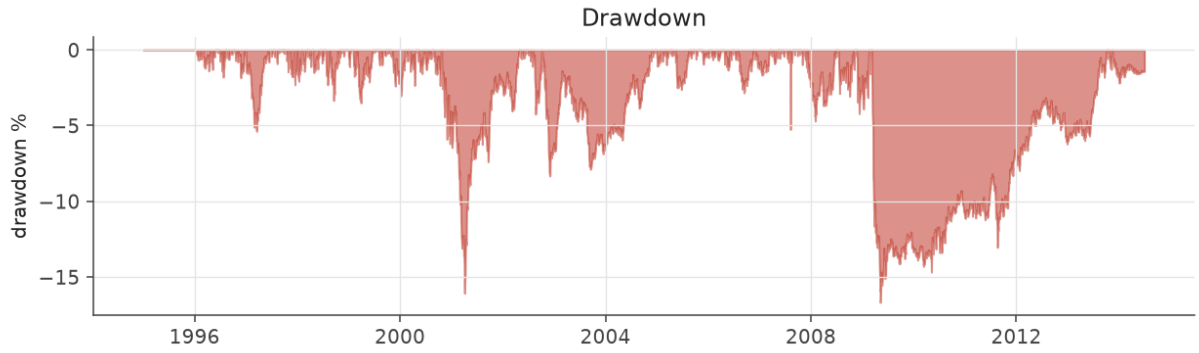
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 1.23

• best return among equity: `market_boost_max` (0.15); this one: 0.0627

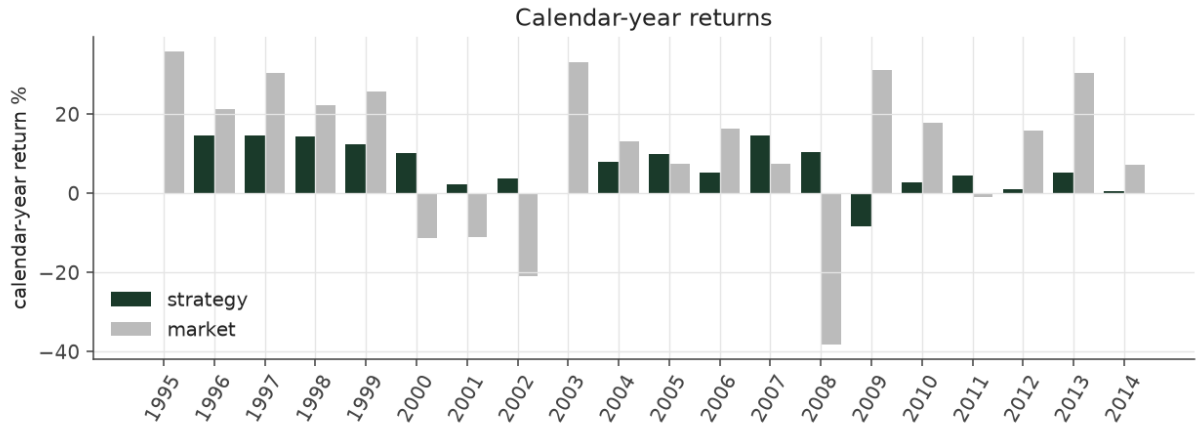
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.376



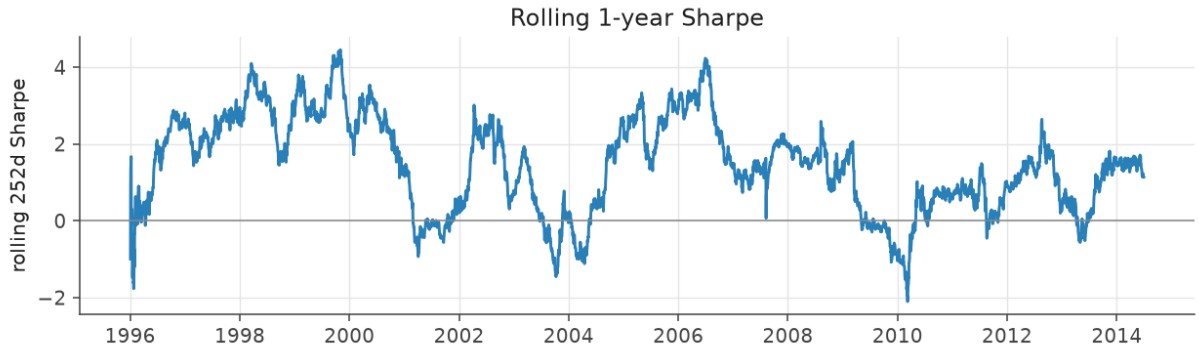
NAV (log scale)



Drawdown from running peak

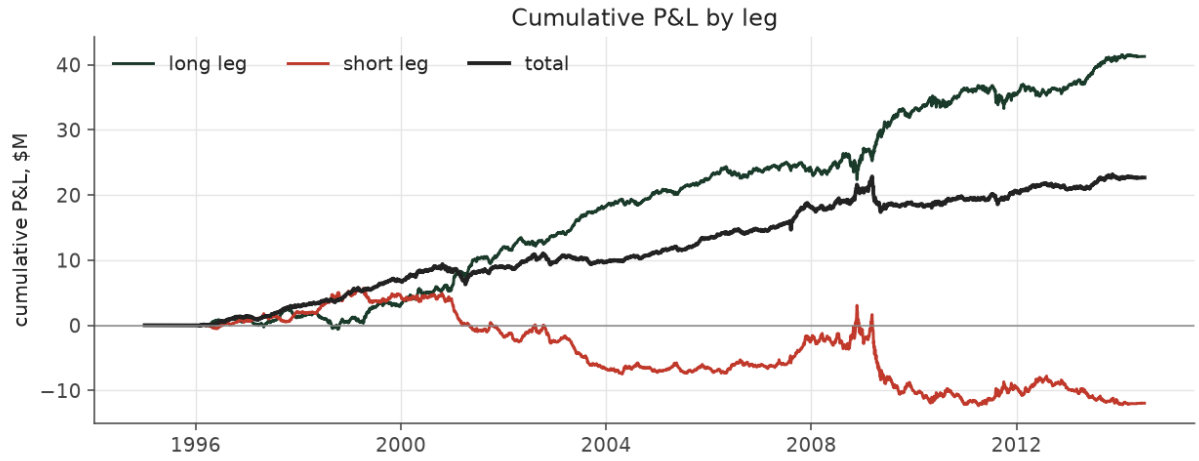


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	41.25	-11.96
P&L after costs ($M)	38.06	-15.40
of which common (average-event) component ($M)	25.96	-26.74
of which excess / PEAD component ($M)	15.29	14.78
excess minus costs ($M)	12.10	11.35
gross return per $ traded (bp)	253.40	-71.65
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.21	-0.26

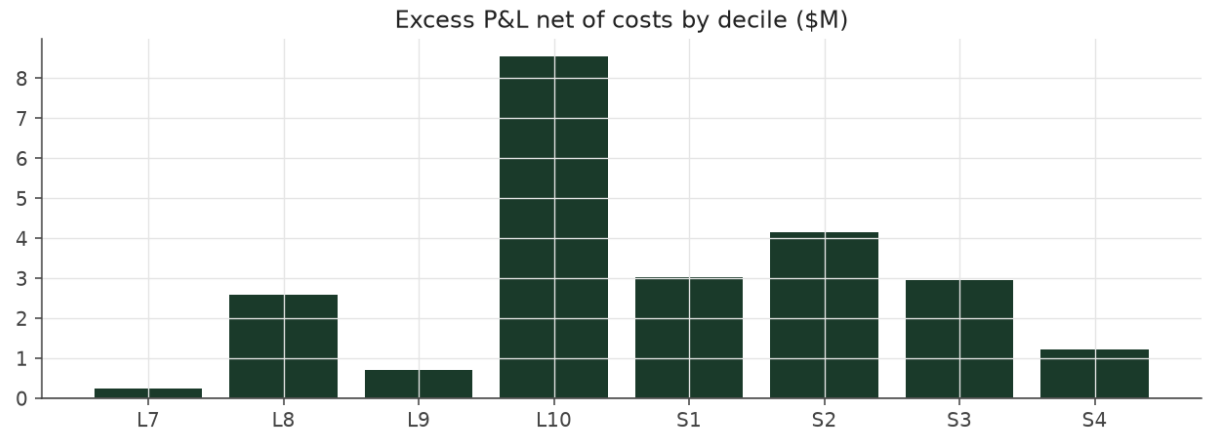
Daily P&L correlation between the two legs: **-0.81**.

• **The short leg loses money in absolute terms** (-12.0M before costs) while the long leg earns +41.3M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 26.0M / 26.7M. Stripping that common component, the surprise signal itself earned +15.3M on the long side and +14.8M on the short side.

• The two legs' daily P&L are -0.81 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.21 long / -0.26 short vs 1.23 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,009	143.22	2.92	0.54	0.28	0.25	1.1%	17.69
long	8	16,508	381.09	8.90	3.35	0.75	2.59	11.1%	67.99
long	9	17,676	504.10	8.05	1.68	0.97	0.70	3.0%	13.98
long	10	17,111	599.56	18.19	9.73	1.18	8.55	36.5%	142.59
short	1	17,093	602.76	-5.93	4.33	1.29	3.04	13.0%	50.38
short	2	17,204	519.78	-4.61	5.21	1.06	4.15	17.7%	79.90
short	3	15,897	390.13	-3.46	3.73	0.79	2.95	12.6%	75.50
short	4	9,154	156.96	-1.39	1.51	0.30	1.21	5.2%	77.08



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,166	300.07	5.43	7.33	1.15	6.18	26.4%	205.99
2	22,785	620.13	5.83	7.93	1.74	6.19	26.4%	99.88
3	25,396	756.47	5.70	7.17	1.58	5.60	23.9%	73.97
4	27,473	807.13	3.43	4.58	1.26	3.32	14.2%	41.11
5	28,832	813.80	2.29	3.05	0.89	2.15	9.2%	26.48

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,618	221.29	5.14	6.66	1.11	5.55	23.7%	250.73
2	23,010	655.74	5.53	8.35	1.97	6.38	27.2%	97.26
3	25,773	804.72	5.97	8.09	1.61	6.48	27.6%	80.53
4	27,539	811.43	4.28	4.77	1.14	3.63	15.5%	44.80

5	28,707	804.24	1.74	2.20	0.80	1.39	5.9%	17.34

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,693	1,250.38	-0.79	6.30	2.20	4.10	17.5%	32.79
41-60d	25,536	745.51	4.38	4.54	1.53	3.02	12.9%	40.45
61-90d	35,163	970.88	6.82	8.31	2.05	6.26	26.7%	64.47
>90d	12,260	330.83	12.25	10.91	0.84	10.07	42.9%	304.33

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,652	695.59	8.69	10.31	1.36	8.95	38.2%	128.69
2002-07	39,722	1,084.81	9.34	11.51	2.13	9.37	40.0%	86.37
2008-13	45,278	1,517.20	4.64	8.25	3.13	5.12	21.8%	33.76

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,907	322.97	1.41	2.09	0.68	1.41	6.0%	43.55
Chems	2,587	220.11	1.30	1.70	0.39	1.31	5.6%	59.38
Durbl	2,836	237.64	2.34	2.85	0.44	2.41	10.3%	101.37
Enrgy	6,186	269.12	3.79	4.39	0.51	3.89	16.6%	144.43
Hlth	11,622	313.38	2.57	3.25	0.66	2.58	11.0%	82.50
Manuf	12,007	294.12	1.73	2.50	0.61	1.89	8.1%	64.36
Money	17,569	288.48	2.51	3.26	0.65	2.61	11.2%	90.62
NoDur	5,074	265.27	1.25	1.74	0.51	1.23	5.3%	46.45
Other	18,309	313.12	2.26	3.06	0.70	2.36	10.1%	75.48
Shops	11,943	306.44	3.05	3.71	0.63	3.08	13.1%	100.58
Telcm	3,942	260.75	-0.03	0.49	0.47	0.01	0.0%	0.44
Utils	4,670	206.21	0.48	1.03	0.38	0.65	2.8%	31.61

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,046	1,043.79	3.54	6.67	1.89	4.78	20.4%	45.74
low vol	40,091	1,128.50	9.89	12.40	2.49	9.91	42.3%	87.81
mid vol	37,903	1,073.74	9.08	10.73	2.13	8.60	36.7%	80.08

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	6.27%	0.00%
0.5%	1.7	5.98%	0.29%
1.0%	3.4	5.67%	0.59%
2.0%	6.8	5.01%	1.26%
5.0%	16.9	2.36%	3.90%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.